Detection of Exoplanet Transits using Convolutional Neural Networks on Simulated Light Curves

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Abstract

Exoplanet detection is an important area of modern astronomy. One of the most widely used methods is the transit method, where a planet passing in front of a star causes a slight dip in its brightness. In this project, a Convolutional Neural Network (CNN) was developed to detect exoplanet transits using simulated light curve data. Synthetic light curves were generated by modeling stellar brightness and introducing artificial dips to represent planetary transits. Gaussian noise and variability were added to simulate realistic observational conditions. The model achieved an accuracy of approximately 0.69 in classifying transit and non-transit signals. This study demonstrates that deep learning techniques can effectively identify exoplanet signals in noisy data.

Introduction

The discovery of exoplanets has significantly expanded our understanding of planetary systems beyond our solar system. One of the most successful techniques for detecting exoplanets is the transit method, where a planet passing in front of its host star causes a temporary decrease in observed brightness.

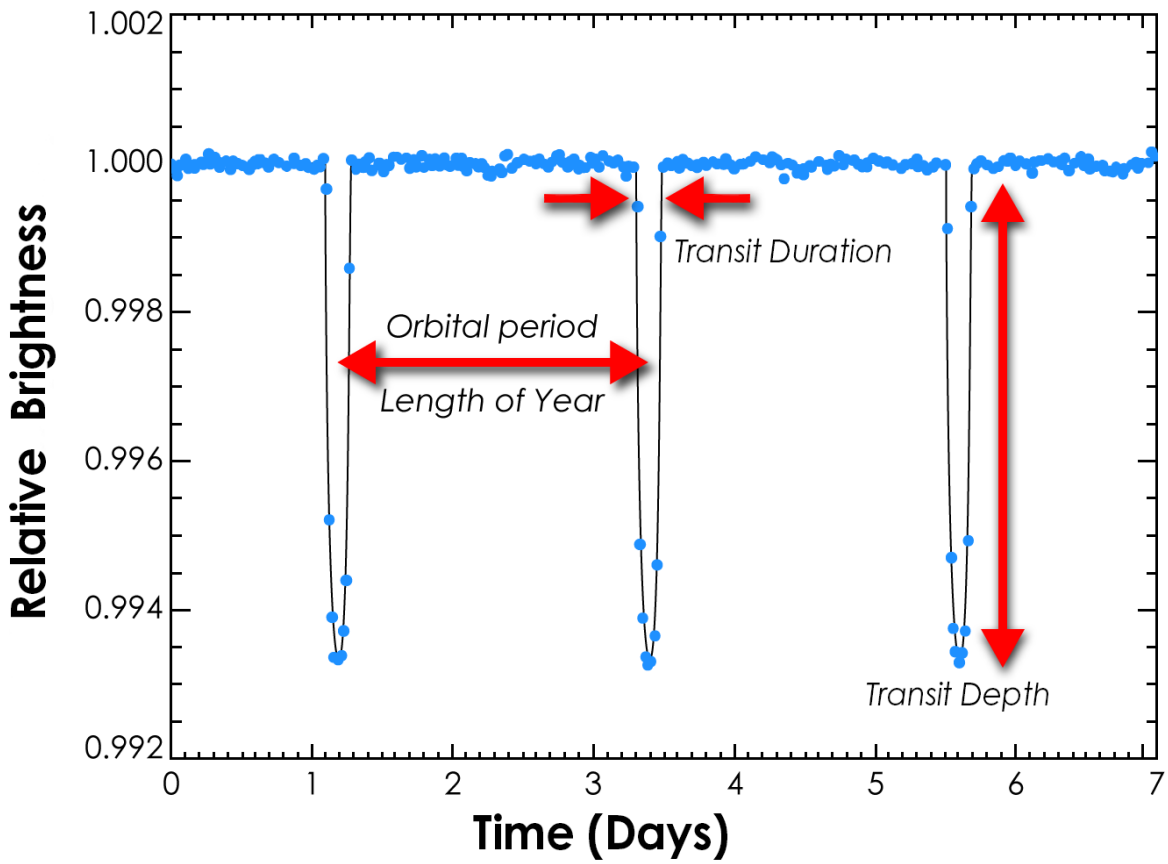
Space missions such as Kepler have used this method to discover thousands of exoplanets. However, detecting these signals is challenging due to noise and variability in stellar data.

With advancements in machine learning, especially deep learning, it has become possible to automate the detection of such patterns. This project aims to simulate light curve data and use a Convolutional Neural Network (CNN) to identify transit signals.

Methodology

- ## Light Curve Simulation

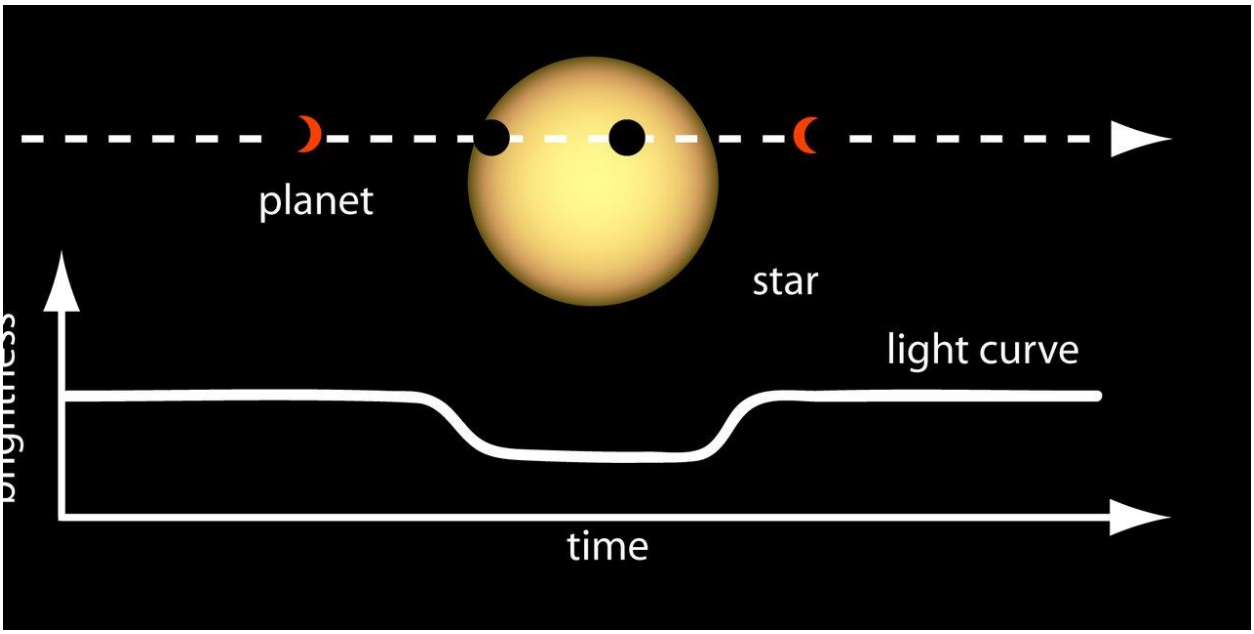
Synthetic light curves were generated to represent the brightness of stars over time. These light curves serve as the input data for the model.



:- Example of a light curve showing brightness variation over time. The dip represents a possible exoplanet transit.

- ## Transit Signal Generation

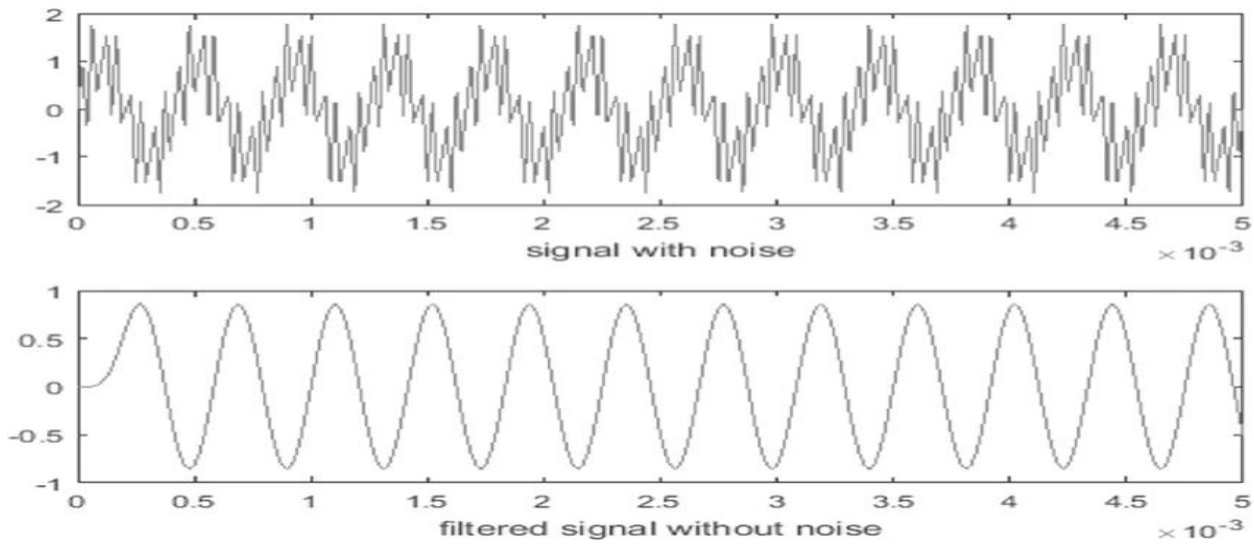
Artificial transit events were created by introducing small dips in brightness at random positions. These dips simulate the effect of a planet passing in front of a star.



:- Illustration of the transit method, where a planet passing in front of a star causes a temporary drop in observed brightness.

- ## Noise and Variability

To make the dataset realistic, Gaussian noise and random fluctuations were added. This mimics real astronomical observations where data is often noisy.



:- Comparison between a clean signal and a noisy light curve. Noise makes transit detection more challenging.

- ### Dataset Preparation

The dataset consisted of labeled samples:

- **1** → Transit present
- **0** → No transit

This allowed the problem to be treated as a binary classification task.

3. Model Architecture

A Convolutional Neural Network (CNN) was used to detect patterns in the light curves.

The model includes:

- Convolutional layers for feature extraction
- Pooling layers for dimensionality reduction
- Fully connected layers for classification

The CNN is effective in identifying local patterns such as dips in brightness.

4. Results

The model was trained on the simulated dataset for multiple epochs. The training loss gradually decreased, indicating that the model was learning meaningful patterns.

The final accuracy achieved was approximately:

Accuracy ≈ 0.67

This shows that the model can detect transit signals even in the presence of noise.

```
(venv) abhishek@abhishek-HP-Notel
/main.py
Epoch 43, Loss: 0.6421
Epoch 44, Loss: 0.6400
Epoch 45, Loss: 0.6379
Epoch 46, Loss: 0.6358
Epoch 47, Loss: 0.6340
Epoch 48, Loss: 0.6320
Epoch 49, Loss: 0.6299
Epoch 50, Loss: 0.6281

🚀 PyTorch Accuracy: 0.67
📈 Showing sample curves...
```

:- Sample simulated light curves showing transit and non-transit signals used for training the model.

5. Discussion

Although the model achieved reasonable accuracy, there are several limitations:

- The dataset is synthetic and may not fully represent real astronomical data
- Noise patterns in real observations can be more complex
- The model may struggle with very small or irregular transit signals

Despite these limitations, the results demonstrate the potential of CNNs in time-series classification tasks related to astronomy.

6. Future Work

Future improvements can include:

- Using real datasets such as TESS or Kepler data
- Improving model architecture for better accuracy
- Detecting multiple planets in a single light curve
- Adding evaluation metrics such as precision and recall

7. Conclusion

This project demonstrated the use of Convolutional Neural Networks for detecting exoplanet transits in simulated light curve data. By generating realistic synthetic data and training a deep learning model, it was possible to identify transit signals with moderate accuracy. This approach highlights the potential of machine learning in modern astrophysics and data-driven discovery.

8. References

1. NASA Exoplanet Archive
2. Kepler Mission Data
3. Basic Machine Learning and Deep Learning resources